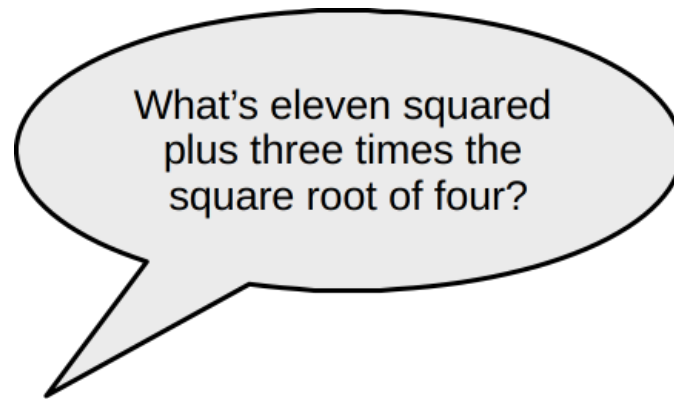


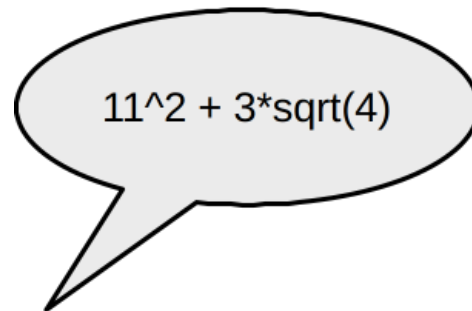
Storing Data in R (Solutions)

Basic data types and structures

R lets us communicate deterministically with a computer.



??!?!?!?



[1] 127



Complex expressions and objects are stored in **variables**.

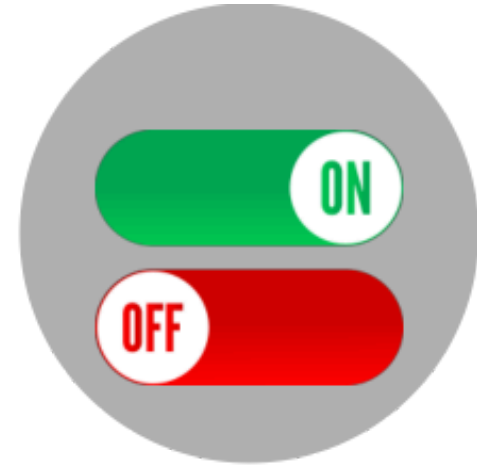
Three basic types:



Numeric



Character



Logical

Exercises

Which type should be used to store these variables?

- Name
- Social security number
- Height
- Highest level of education
- If someone voted in the last election
- Whether someone is married



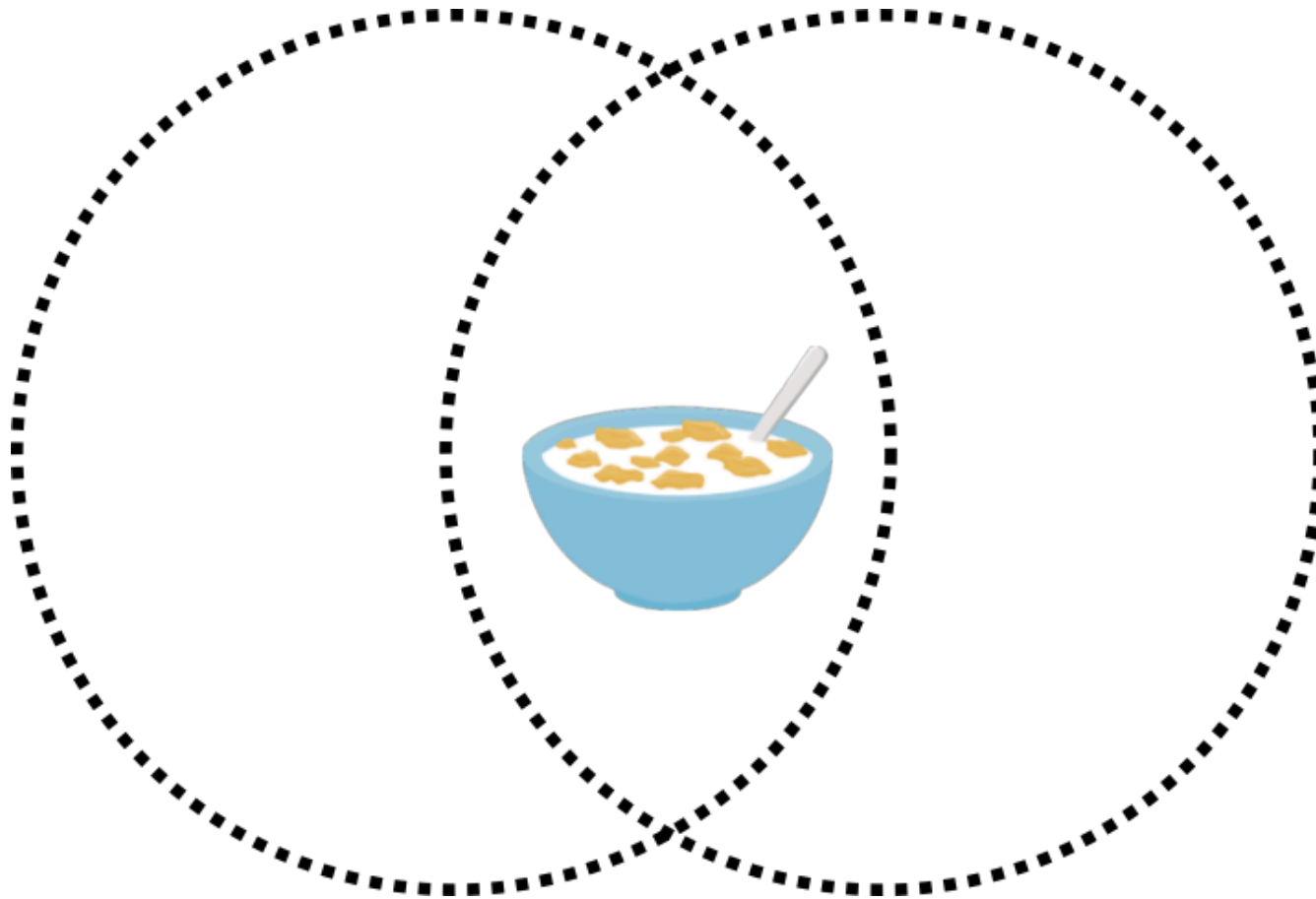
Solutions/Hints

Some answers may not be super obvious, and some variables may have more than one potential answer.

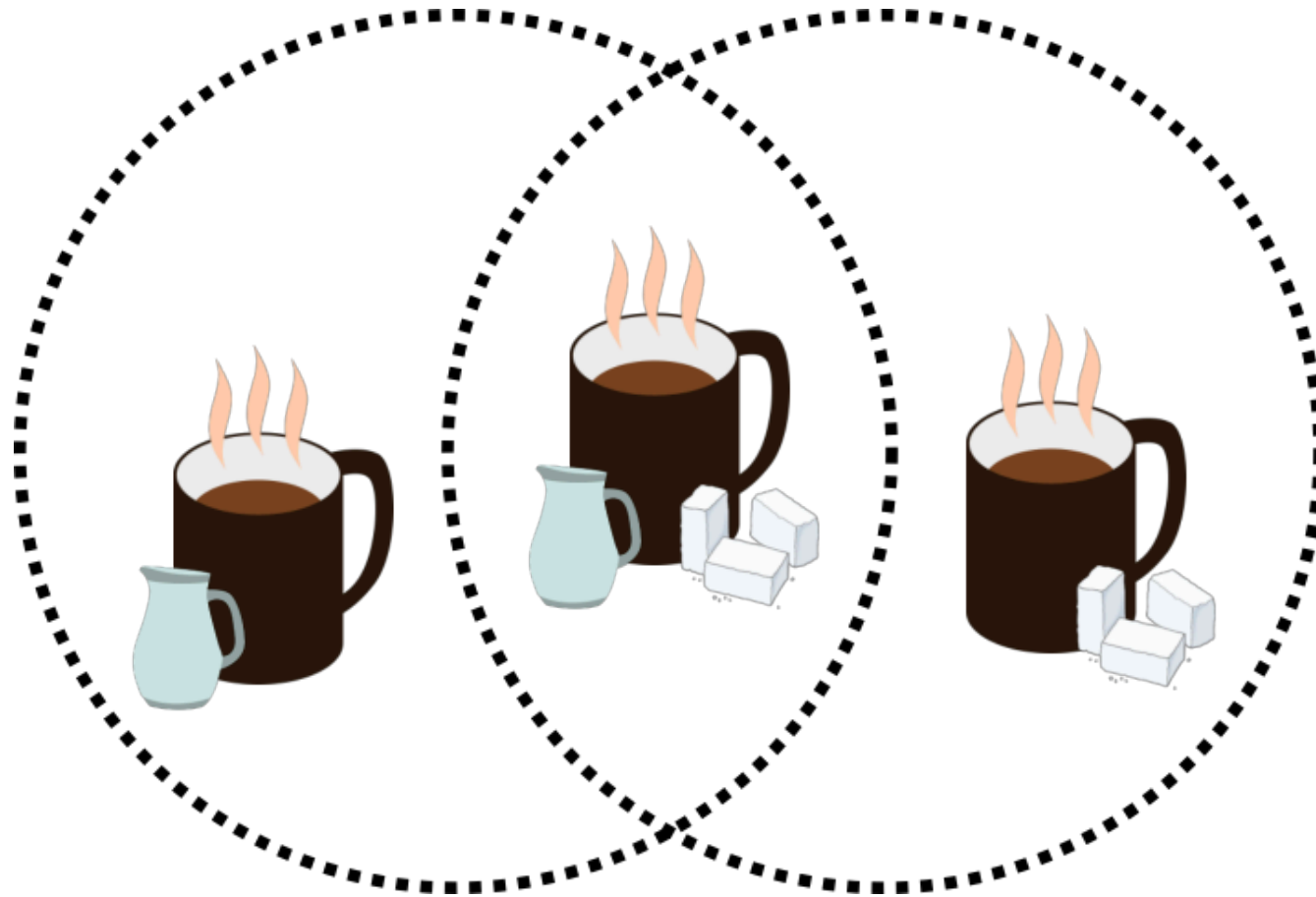
Logical operators return a logical value.

- `<`, `>`, `<=`, `>=` to compare size
- `==` or `!=` for exact equality
- `&` for “and”,
 `|` for “or”





& checks if both sides are **TRUE**



| checks if at least one side is **TRUE**

SECRET THIRD THING: NA

- Not 0, not NaN
- Needs its own special set of functions



We typically work with a lot of data, not just a single object. R uses **vectorization** to handle data in bulk.

Ex: Height conversions

The first person in our data is 68 inches tall → 173 cm.

The next person in our data is 63 inches tall → 160 cm.

Inches → Multiply by 2.54 → Round → Centimeters

Data of the same type is stored in a **vector**.

1	2	3	4	5
68	63	70	67	61

Operations are done to each element.

1	2	3	4	5	to cm →	1	2	3	4	5
68	63	70	67	61		173	160	178	170	155

Exercises

Vectorize the code for each prompt.

- Convert the course names to uppercase.
- For what values of x is $[(\frac{1}{2}x - 8)^2 - 20]$ negative?
- CHALLENGE: Vectorize the **for** loop that creates a meal plan.



Solutions/Hints

- The `toupper ( )` function can have a character vector as input.
- Start with a vector of the numbers 0-50.
- The `paste ( )` function can take multiple vectors as input.
How can we get the numbers to order each meal?



To manually define a vector, use `c()`

For specific numeric vectors, use `:` or `seq()`

Vectors can also be generated with a more involved process.

The `replicate()` command is useful for generating data.

`replicate(n, expression)`

↑
number
of times

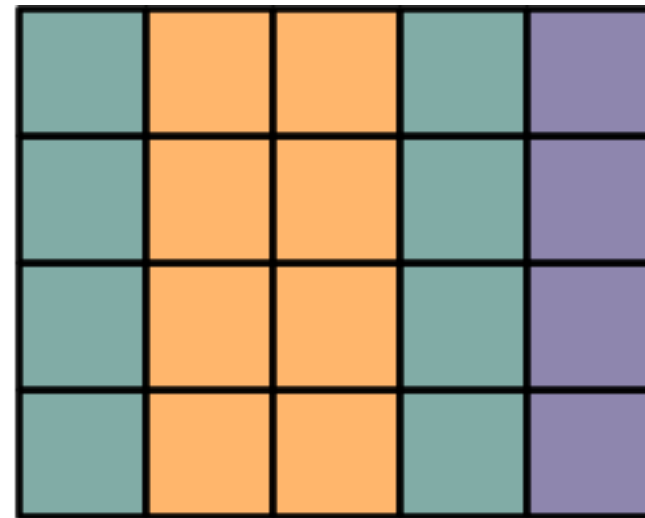
↑
operation
to repeat

Vectors store one type of data - how to record more than one measurement per object?

Dataframes!



Vector



Dataframe

Items are *rows*, variables are *columns*.

Use `tibble()` to build a dataframe by hand.

Or use `read_csv()` to load data from a specific location.



```
read_csv(data location)
```

↑
must be
exact!!

Looking ahead

In [chapter 4](#), we'll learn how to summarize data visually and numerically. If you want a head start, run the following code to make a histogram of (fake) emergency room wait times:

```
1 library(tidyverse)
2 wait_times <- tibble(hours = c(1.3, 0.7, 3.2, 5.5, 1.0,
3                                0.1, 2.3, 1.5, 1.6, 2.6,
4                                0.8, 0.5, 1.4, 1.7, 1.9))
5
6 ggplot(wait_times, aes(x = hours)) +
7   geom_histogram(bins = 10)
```

Do you think the *mean* or *median* ER wait time is longer? How can you tell without the exact calculations?